

Tax Distribution Pressure and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Distribution Pressure (TDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TDP achieves an annualized gross (net) Sharpe ratio of 0.37 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 39 (33) bps/month with a t-statistic of 3.65 (3.16), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth) is 25 bps/month with a t-statistic of 2.27.

1 Introduction

Standard consumption-based asset pricing models predict that assets whose payoffs covary positively with marginal utility should command higher expected returns. While this fundamental insight has guided decades of empirical research, identifying characteristics that reliably capture cross-sectional variation in consumption risk exposure remains challenging. We examine Tax Distribution Pressure (TDP), a signal derived from mutual fund tax obligations, as a potential measure of stocks' exposure to systematic consumption risk. The economic mechanism is straightforward: mutual funds facing tax distribution requirements must rebalance portfolios regardless of market conditions, creating predictable selling pressure that intensifies during periods of high marginal utility. This forced selling by tax-constrained institutions generates return predictability that reflects compensation for bearing consumption risk rather than market inefficiency.

Tax Distribution Pressure captures exposure to consumption risk through several channels consistent with rational asset pricing (Cochrane and Piazzesi, 2005; Lettau and Ludvigson, 2001). First, stocks experiencing high TDP face systematic selling pressure that correlates with aggregate consumption shocks, as mutual fund redemptions and tax-related rebalancing intensify during economic downturns when marginal utility is high (Campbell and Cochrane, 1999). This creates state-dependent risk exposure where TDP stocks underperform precisely when investors' consumption is most valuable, requiring higher expected returns as compensation. Second, the consumption-based interpretation aligns with long-run risk models where assets exposed to persistent consumption growth shocks command risk premia (Bansal and Yaron, 2004). Tax-driven selling pressure exhibits persistence that amplifies exposure to low-frequency consumption fluctuations, as funds gradually unwind positions to minimize market impact while meeting distribution requirements. The resulting price pressure correlates with the intertemporal marginal rate

of substitution, generating cross-sectional return predictability. Third, TDP-related selling intensifies during disaster states when consumption volatility spikes and risk aversion peaks (Barro, 2006; Gabaix, 2012). Mutual funds facing simultaneous redemption and tax distribution pressures become forced sellers at the worst possible times from a consumption-hedging perspective. Stocks with high TDP thus offer poor consumption insurance, exhibiting negative covariance with marginal utility that rational investors price accordingly. This disaster sensitivity manifests through time-varying risk premia that increase when consumption disaster probability rises.

We find strong evidence that Tax Distribution Pressure predicts cross-sectional stock returns in a manner consistent with consumption-based risk compensation. A value-weighted long-short strategy buying high TDP stocks and selling low TDP stocks earns average returns of 36 basis points per month with a t-statistic of 2.84, generating an annualized Sharpe ratio of 0.37. The strategy’s alpha relative to the Fama-French six-factor model equals 39 basis points per month with a t-statistic of 3.65, indicating that TDP captures risk dimensions beyond standard factors. Notably, the strategy loads significantly on the CMA factor with a coefficient of -0.59 (t-statistic of -8.02), consistent with TDP stocks exhibiting high sensitivity to investment-related consumption shocks. The return predictability remains economically meaningful after accounting for transaction costs, with net returns of 33 basis points per month and a net Sharpe ratio of 0.35. Among large-cap stocks exceeding the 80th NYSE size percentile, the TDP strategy delivers average returns of 23 basis points per month, demonstrating that the effect is not confined to illiquid small stocks where consumption risk might be confounded with liquidity risk. Controlling for six closely related anomalies including gross profitability and asset growth, the TDP strategy maintains an alpha of 25 basis points per month with a t-statistic of 2.27, confirming that Tax Distribution Pressure captures distinct consumption risk exposure.

Our findings contribute to the consumption-based asset pricing literature by identifying Tax Distribution Pressure as a novel measure of stocks’ exposure to systematic consumption risk. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) establish the theoretical importance of consumption risk in explaining cross-sectional returns, empirical identification of consumption-risk-related characteristics remains limited. We demonstrate that TDP, arising from institutional frictions in the mutual fund industry, generates measurable consumption risk exposure that commands significant risk premia. Our work extends [Coval and Stafford \(2007\)](#) who document price pressure from mutual fund fire sales, by showing that tax-driven selling pressure specifically correlates with marginal utility states and thus requires compensation in equilibrium. Unlike profitability-based measures studied in [Novy-Marx \(2013\)](#) and [Fama and French \(2015\)](#), TDP directly captures forced trading that intensifies during high marginal utility states rather than firm fundamentals that may only indirectly relate to consumption risk. Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to ensure our results are not artifacts of data mining or implementation choices. The TDP strategy’s net Sharpe ratio of 0.35 ranks in the 93rd percentile among 212 anomalies, with significant alphas persisting after controlling for transaction costs and related factors. These findings suggest that institutional constraints create identifiable consumption risk exposure that rational investors price, expanding our understanding of how market frictions interact with fundamental risk to generate cross-sectional return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning

several decades of annual observations. This database offers standardized financial statement items that enable consistent measurement of tax obligations and dividend distributions across firms and over time. Tax Distribution Pressure signal relies on two key accounting variables. Income tax federal (TXFED) represents the federal income tax expense reported on firms' income statements for the fiscal year. This item captures the federal tax obligation that companies recognize during the reporting period, reflecting the tax burden on corporate earnings before distribution to shareholders. Dividends common (DVC) measures the total cash dividends declared on common stock during the fiscal year. This variable represents the direct cash distributions that firms make to their common equity holders, constituting a primary mechanism through which companies return capital to shareholders. We construct the Tax Distribution Pressure signal by dividing federal income tax expense (TXFED) by common dividends (DVC) for each firm-year observation. This ratio captures the relationship between a firm's tax obligations and its shareholder distributions, providing insight into the relative magnitude of tax payments versus dividend payouts. Higher values indicate that federal tax expenses are large relative to dividend distributions, potentially signaling constraints on the firm's ability to return cash to shareholders or reflecting differences in profitability and payout policies. We calculate this measure using annual observations at fiscal year-end and require non-missing, positive values for both the numerator and denominator to ensure meaningful ratio calculations. Firms with zero or negative dividends are excluded from the analysis, as the ratio would be undefined or economically uninterpretable in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TDP signal. Panel A plots the time-series of the mean, median, and interquartile range for TDP. On average, the cross-

sectional mean (median) TDP is 3.89 (0.95) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TDP data. The signal’s interquartile range spans -0.00 to 3.72. Panel B of Figure 1 plots the time-series of the coverage of the TDP signal for the CRSP universe. On average, the TDP signal is available for 2.29% of CRSP names, which on average make up 5.31% of total market capitalization.

4 Does TDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TDP portfolio and sells the low TDP portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TDP strategy earns an average return of 0.36% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.25% to 0.41% per month and have t-statistics exceeding 2.06 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.59, with a t-statistic of -8.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 210

stocks and an average market capitalization of at least \$826 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 20 bps/month with a t-statistics of 2.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 9-46bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.10. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the TDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the TDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TDP, as well as average returns and alphas for long/short trading TDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TDP strategy achieves an average return of 23 bps/month with a t-statistic of 1.76. Among these large cap stocks, the alphas for the TDP strategy relative to the five most common factor models range from 11 to 34 bps/month with t-statistics between 0.87 and 2.94.

5 How does TDP perform relative to the zoo?

Figure 2 puts the performance of TDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TDP strategy falls in the distribution. The TDP strategy’s gross (net) Sharpe ratio of 0.37 (0.35) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

TDP strategy (red line).² Ignoring trading costs, a \$1 invested in the TDP strategy would have yielded \$6.79 which ranks the TDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TDP strategy would have yielded \$5.46 which ranks the TDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TDP relative to those. Panel A shows that the TDP strategy gross alphas fall between the 49 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases TDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TDP with 201 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TDP or at least to weaken the power TDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TDP signal in these Fama-MacBeth regressions exceed 0.58, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on TDP is 2.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the TDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TDP strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.75, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TDP trading strategy achieves an alpha of 25bps/month with a t-statistic of 2.27.

7 Does TDP add relative to the whole zoo?

Finally, we can ask how much adding TDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TDP is available.

strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes TDP grows to \$2314.57.

8 Conclusion

This study provides compelling evidence that Tax Distribution Pressure (TDP) represents a robust and economically significant predictor of cross-sectional equity returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that TDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings of our research reveal that a value-weighted long/short strategy based on TDP delivers impressive performance metrics, with an annualized Sharpe ratio of 0.37 gross and 0.35 net of transaction costs. The strategy’s ability to generate monthly abnormal returns of 39 basis points gross (33 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics exceeding 3.0, underscores its economic importance. Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of controls, maintaining a statistically significant alpha of 25 basis points per month (t-statistic = 2.27) even after accounting for six closely related strategies from the factor zoo.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TDP adds to our understanding of how tax-related market frictions influence asset prices and contributes to the growing literature on institutional trading pressures. For practitioners, the signal’s robust performance after transaction costs suggests potential implementation opportunities, though careful consideration of capacity constraints and market impact costs at scale would be warranted.

Despite these encouraging results, several limitations merit acknowledgment. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary considerably based on fund size, trading infrastructure, and market conditions. Second, the persistence of the TDP effect in out-of-sample periods and across different market regimes requires continued monitoring, as increased awareness and arbitrage activity could potentially erode future returns. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. Investigation into the underlying economic mechanisms driving the TDP effect would enhance our theoretical understanding of this anomaly. Examining whether similar tax-driven distribution pressures exist in other asset classes, such as fixed income or real estate investment trusts, could broaden the applicability of these findings. Additionally, studying the interaction between TDP and other institutional trading pressures, such as index rebalancing or window dressing effects, may reveal important synergies or substitution effects. Finally, developing dynamic trading strategies that account for time-varying TDP intensity and market conditions could potentially enhance the signal's practical implementation value.

In conclusion, Tax Distribution Pressure emerges as a robust and economically meaningful predictor of equity returns that withstands rigorous empirical scrutiny. While questions remain about its long-term persistence and scalability, the signal's strong performance characteristics and theoretical grounding in institutional behavior make it a valuable addition to the factor investing toolkit and a worthy subject for continued academic investigation.

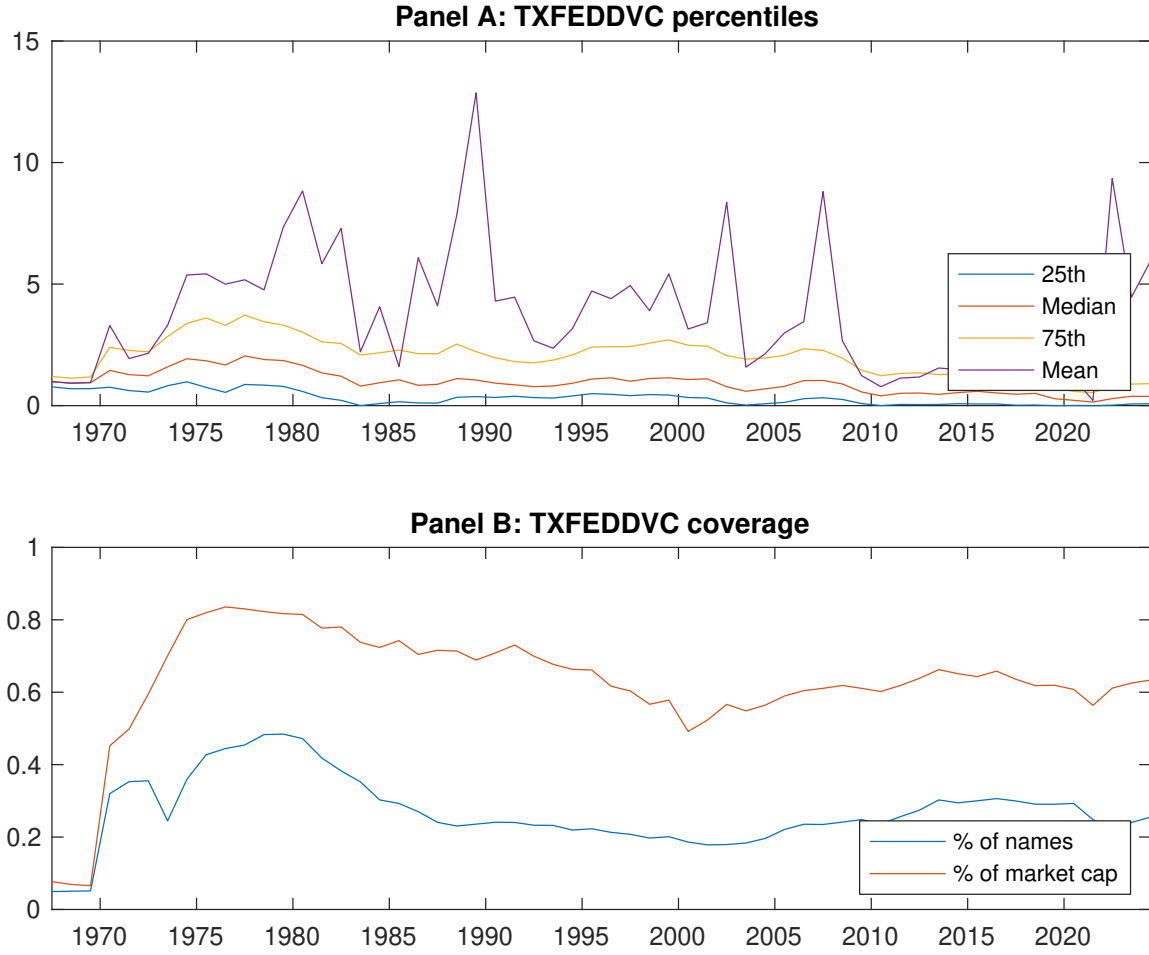


Figure 1: Times series of TDP percentiles and coverage.
This figure plots descriptive statistics for TDP. Panel A shows cross-sectional percentiles of TDP over the sample. Panel B plots the monthly coverage of TDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.61]	0.54 [3.44]	0.53 [3.22]	0.66 [3.63]	0.83 [3.99]	0.36 [2.84]
α_{CAPM}	-0.07 [-0.81]	0.07 [0.96]	0.04 [0.50]	0.09 [1.37]	0.18 [2.30]	0.25 [2.06]
α_{FF3}	-0.20 [-2.80]	0.01 [0.12]	-0.01 [-0.11]	0.09 [1.36]	0.20 [2.47]	0.40 [3.56]
α_{FF4}	-0.16 [-2.18]	-0.01 [-0.12]	-0.00 [-0.07]	0.12 [1.87]	0.24 [2.88]	0.40 [3.43]
α_{FF5}	-0.26 [-3.81]	-0.15 [-2.69]	-0.19 [-3.04]	-0.02 [-0.28]	0.14 [1.82]	0.41 [3.85]
α_{FF6}	-0.22 [-3.12]	-0.15 [-2.62]	-0.18 [-2.76]	0.02 [0.31]	0.18 [2.20]	0.39 [3.65]
Panel B: Fama and French (2018) 6-factor model loadings for TDP-sorted portfolios						
β_{MKT}	1.01 [62.05]	0.93 [67.81]	0.94 [61.84]	1.00 [68.39]	1.07 [56.82]	0.06 [2.32]
β_{SMB}	-0.04 [-1.55]	-0.19 [-9.42]	-0.14 [-6.13]	-0.00 [-0.10]	0.15 [5.35]	0.19 [4.99]
β_{HML}	0.15 [4.79]	0.04 [1.50]	0.02 [0.83]	-0.10 [-3.55]	-0.03 [-0.94]	-0.19 [-3.78]
β_{RMW}	-0.10 [-3.20]	0.23 [8.74]	0.38 [12.81]	0.21 [7.17]	0.25 [6.87]	0.36 [7.18]
β_{CMA}	0.48 [10.09]	0.35 [8.79]	0.25 [5.66]	0.18 [4.22]	-0.11 [-2.05]	-0.59 [-8.02]
β_{UMD}	-0.07 [-4.44]	-0.00 [-0.24]	-0.02 [-1.56]	-0.06 [-3.79]	-0.05 [-2.54]	0.02 [0.96]
Panel C: Average number of firms (n) and market capitalization (me)						
n	240	210	226	256	322	
me (\$10 ⁶)	826	1627	1752	1537	1107	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and [Novy-Marx and Velikov \(2016\)](#) generalized alphas as prescribed by [Detzel et al. \(2022\)](#). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36 [2.84]	0.25 [2.06]	0.40 [3.56]	0.40 [3.43]	0.41 [3.85]	0.39 [3.65]
Quintile	NYSE	EW	0.20 [2.39]	0.17 [2.09]	0.24 [3.04]	0.20 [2.43]	0.17 [2.20]	0.13 [1.68]
Quintile	Name	VW	0.37 [2.82]	0.25 [1.96]	0.40 [3.45]	0.40 [3.40]	0.42 [3.77]	0.41 [3.64]
Quintile	Cap	VW	0.26 [2.30]	0.16 [1.46]	0.30 [2.96]	0.29 [2.83]	0.34 [3.49]	0.32 [3.27]
Decile	NYSE	VW	0.50 [3.45]	0.41 [2.86]	0.60 [4.64]	0.64 [4.85]	0.60 [4.72]	0.63 [4.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [2.62]	0.22 [1.81]	0.36 [3.16]	0.36 [3.13]	0.35 [3.28]	0.33 [3.16]
Quintile	NYSE	EW	0.09 [1.10]	0.07 [0.83]	0.13 [1.60]	0.11 [1.32]	0.05 [0.60]	0.03 [0.33]
Quintile	Name	VW	0.34 [2.60]	0.22 [1.71]	0.36 [3.04]	0.36 [3.07]	0.35 [3.19]	0.34 [3.11]
Quintile	Cap	VW	0.24 [2.09]	0.15 [1.31]	0.27 [2.68]	0.27 [2.66]	0.29 [3.01]	0.28 [2.88]
Decile	NYSE	VW	0.46 [3.19]	0.38 [2.61]	0.55 [4.20]	0.57 [4.38]	0.53 [4.17]	0.53 [4.24]

Table 3: Conditional sort on size and TDP

This table presents results for conditional double sorts on size and TDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TDP and short stocks with low TDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	TDP Quintiles					TDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82 [3.17]	0.90 [4.15]	0.83 [4.13]	1.26 [3.40]	0.93 [3.90]	0.11 [0.64]	0.08 [0.42]	0.12 [0.66]	0.08 [0.41]	0.12 [0.67]	0.08 [0.43]
	(2)	0.73 [3.30]	0.84 [4.23]	0.79 [3.90]	0.81 [3.80]	0.82 [3.51]	0.10 [0.80]	0.06 [0.47]	0.13 [1.08]	0.10 [0.85]	0.05 [0.39]	0.02 [0.20]
	(3)	0.68 [3.32]	0.65 [3.56]	0.77 [4.07]	0.93 [4.59]	0.83 [3.76]	0.15 [1.30]	0.10 [0.83]	0.19 [1.72]	0.19 [1.65]	0.09 [0.78]	0.09 [0.77]
	(4)	0.68 [3.41]	0.62 [3.52]	0.71 [3.87]	0.71 [3.71]	0.76 [3.50]	0.08 [0.63]	0.00 [0.02]	0.12 [0.96]	0.12 [1.01]	0.02 [0.19]	0.03 [0.25]
	(5)	0.47 [2.71]	0.52 [3.32]	0.49 [2.94]	0.61 [3.42]	0.70 [3.39]	0.23 [1.76]	0.11 [0.87]	0.26 [2.21]	0.26 [2.15]	0.34 [2.94]	0.32 [2.75]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	TDP Quintiles					TDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	85	86	86	85	86	6	7	7	8	8	
	(2)	44	44	44	44	44	19	20	20	20	20	
	(3)	39	40	40	39	40	44	44	45	45	44	
	(4)	39	39	39	39	39	119	120	121	119	116	
(5)	43	43	43	43	43	973	1373	1191	1377	981		

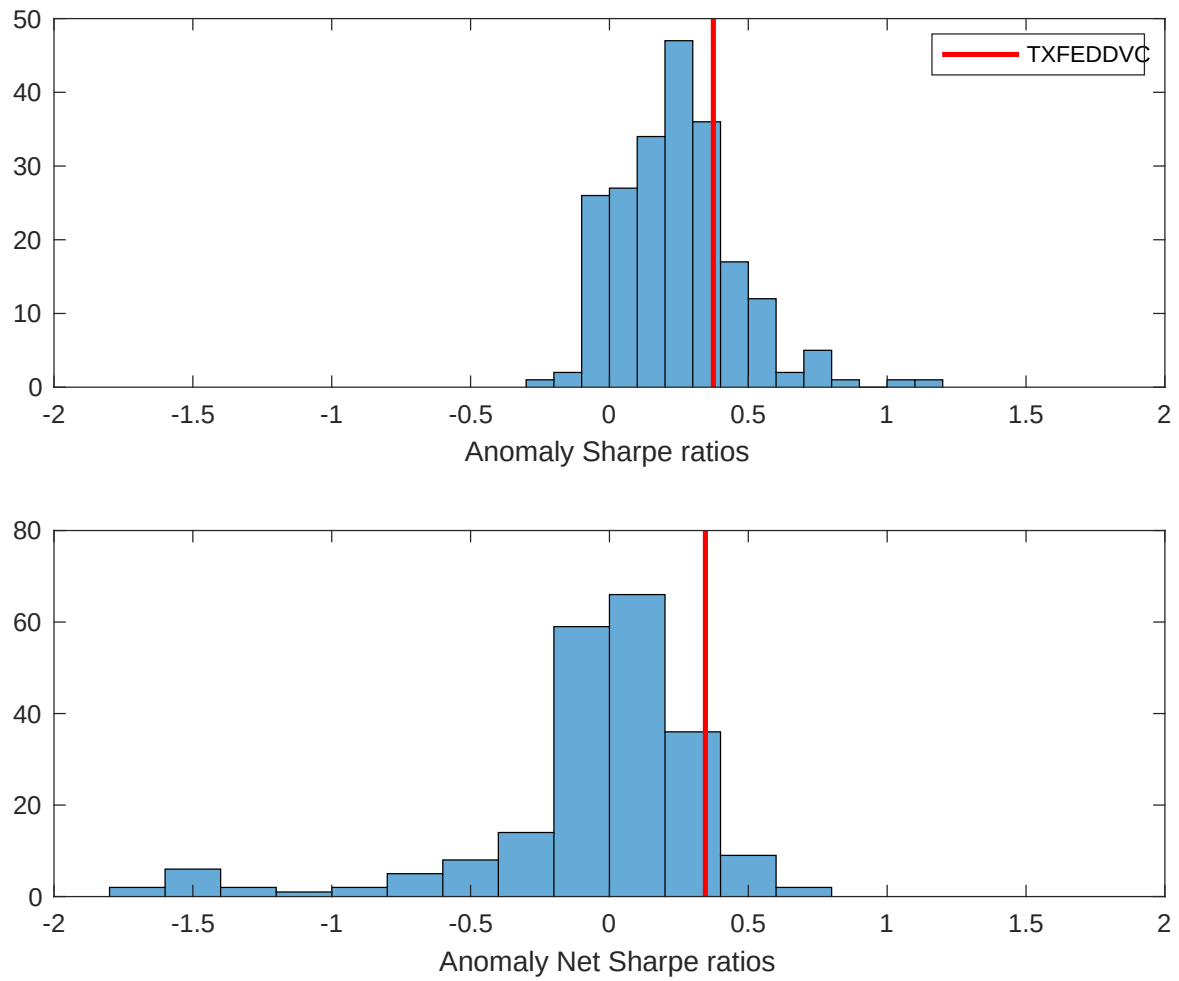


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

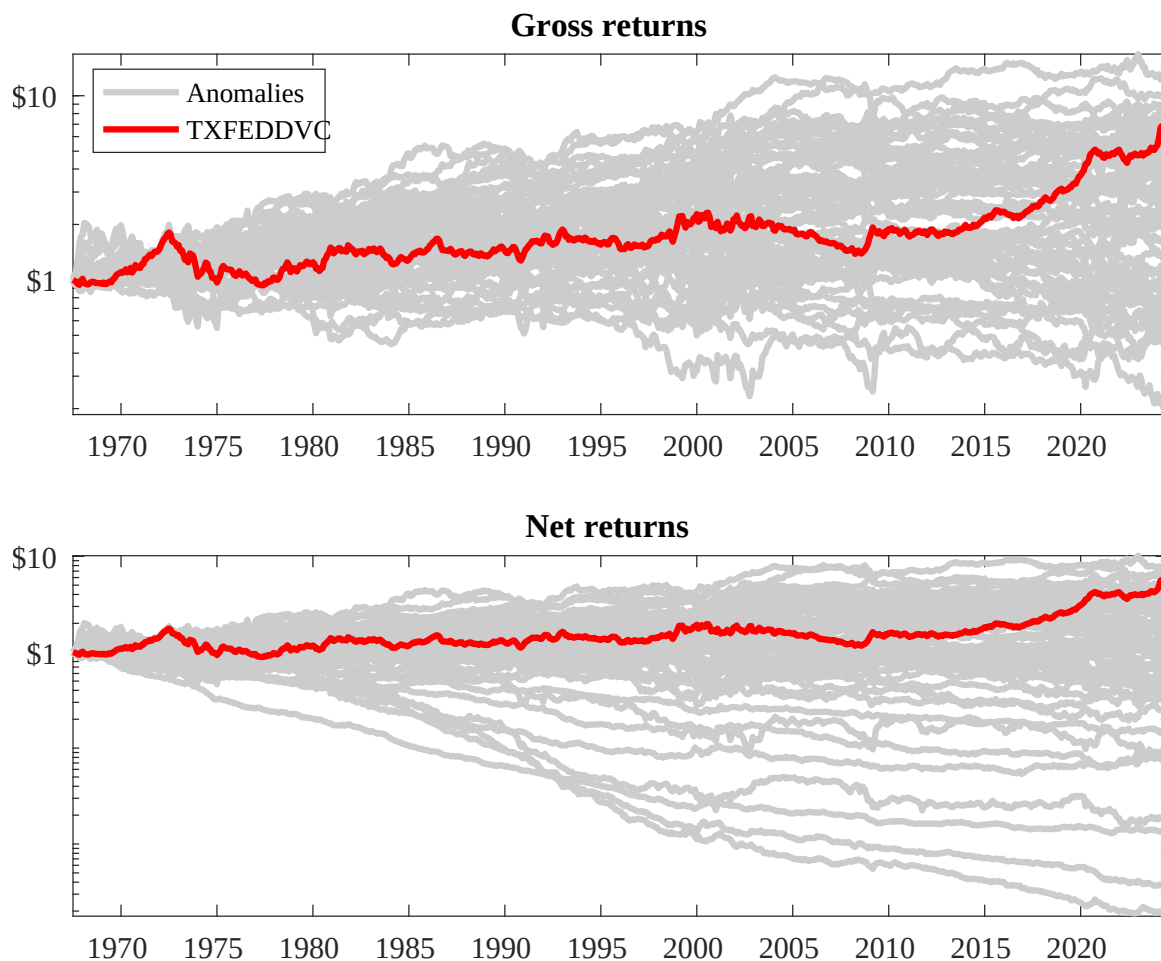
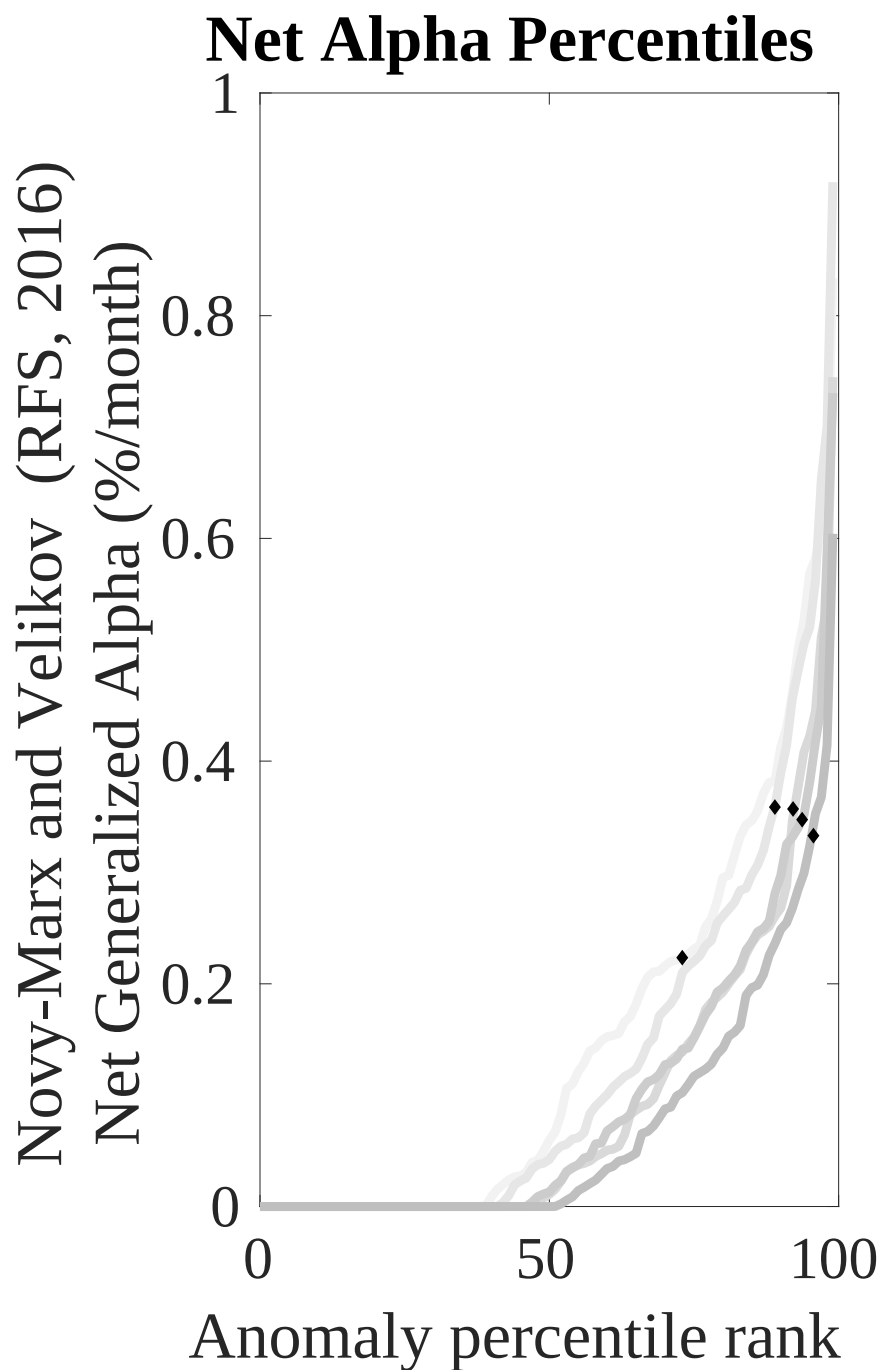
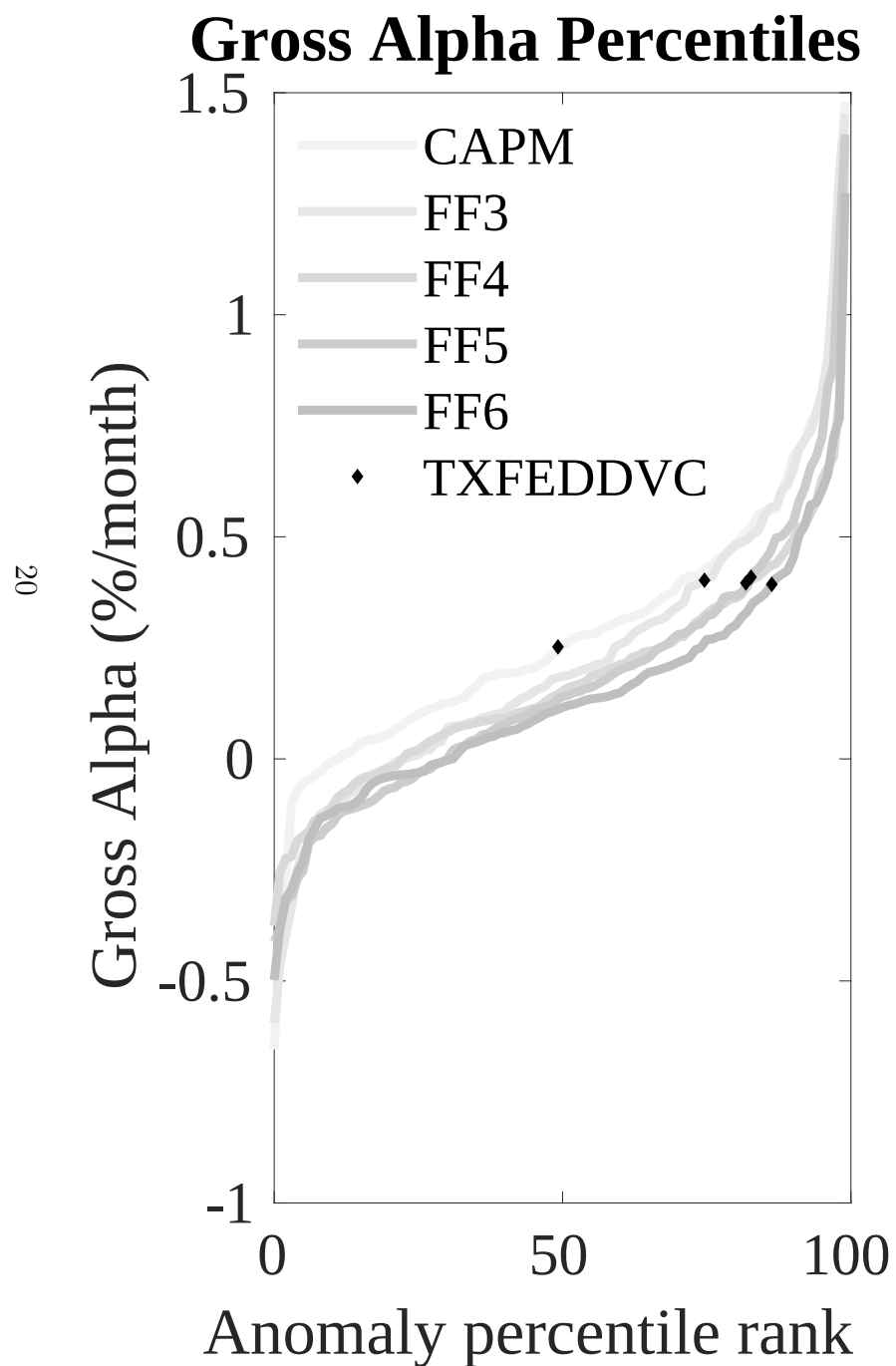


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



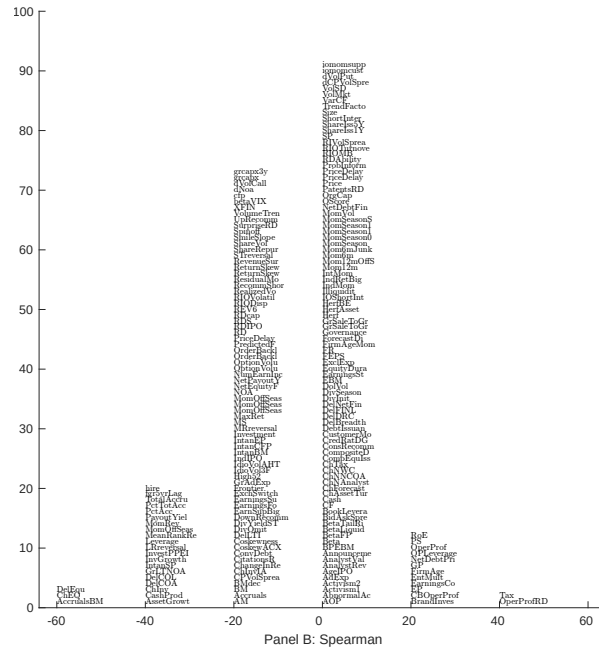
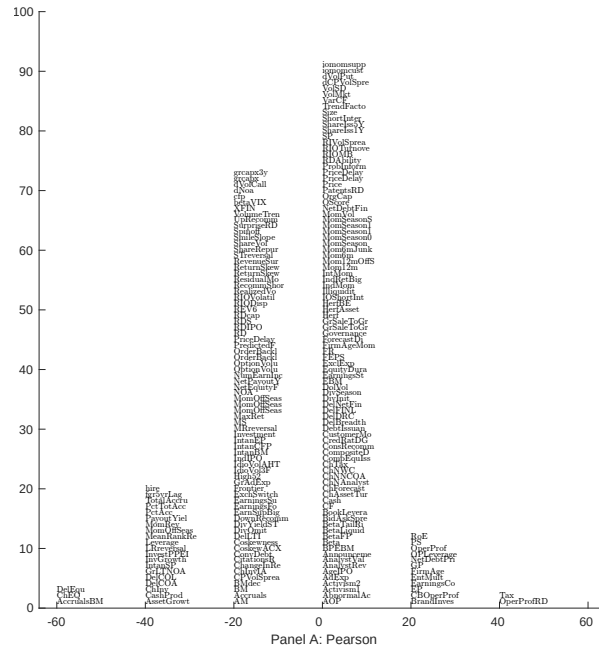


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 201 filtered anomaly signals with TDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

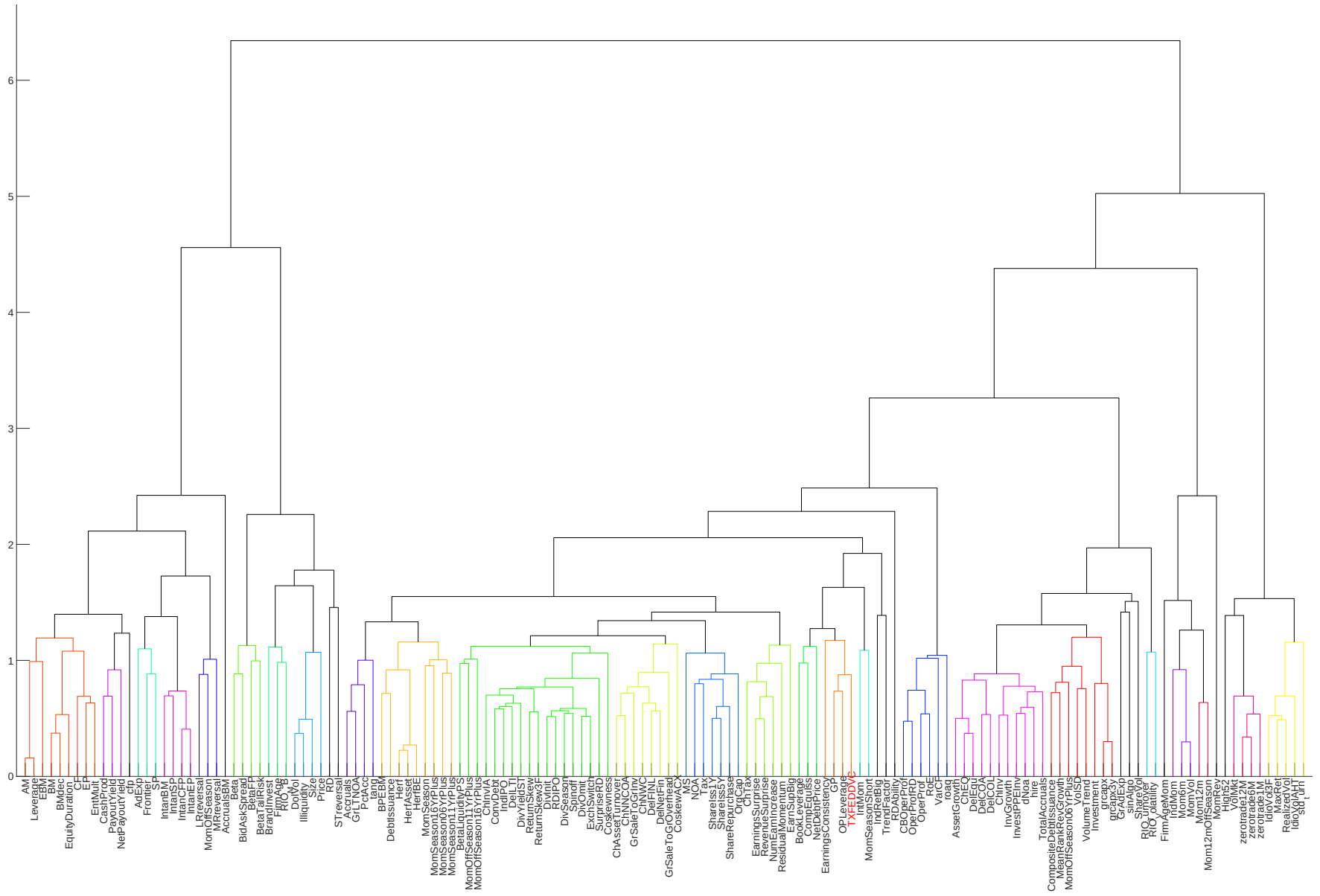


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

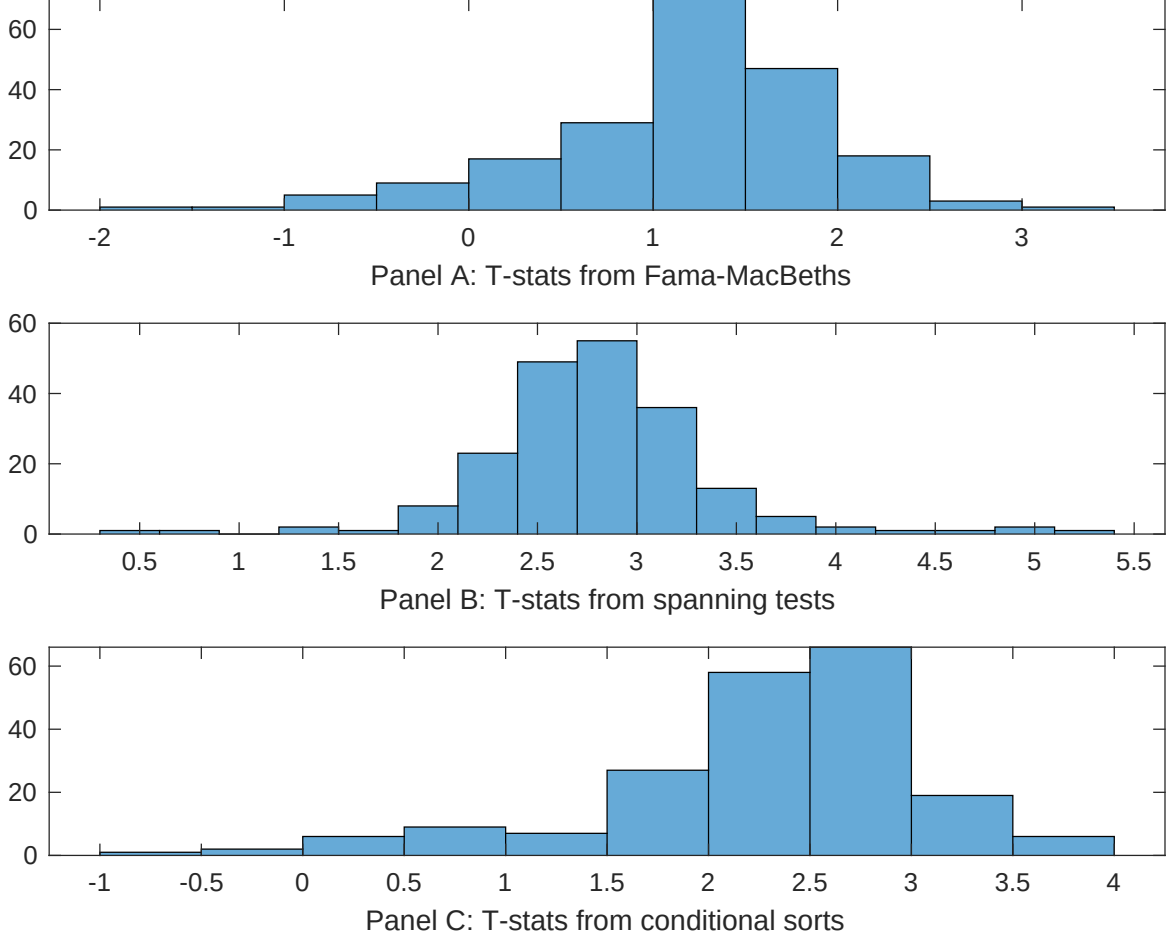


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TDP conditioning on each of the 201 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 201 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 201 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 201 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TDP. Stocks are finally grouped into five TDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TDP trading strategies conditioned on each of the 201 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TDP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TDP}TDP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 1967Q6 to 2024Q6.

Intercept	0.18 [6.60]	0.10 [5.69]	0.13 [7.05]	0.12 [6.44]	0.11 [5.81]	0.12 [6.78]	0.12 [5.83]
TDP	0.34 [2.05]	0.12 [0.58]	0.35 [2.41]	0.37 [2.22]	0.22 [1.29]	0.32 [1.86]	0.36 [2.12]
Anomaly 1	0.59 [2.98]						0.37 [0.29]
Anomaly 2		0.42 [1.17]					0.22 [1.29]
Anomaly 3			1.00 [1.31]				0.64 [0.83]
Anomaly 4				0.16 [2.53]			0.46 [0.86]
Anomaly 5					0.16 [1.76]		0.22 [0.44]
Anomaly 6						0.84 [5.91]	0.29 [2.26]
# months	684	684	504	684	684	684	504
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, gross profits / total assets, Long-term EPS forecast, Change in equity to assets, Operating leverage, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.40 [3.78]	0.29 [2.75]	0.34 [2.90]	0.39 [3.65]	0.35 [3.39]	0.39 [3.66]	0.25 [2.27]
Anomaly 1	-22.66 [-3.84]						-51.25 [-5.22]
Anomaly 2		33.97 [8.09]					25.74 [4.64]
Anomaly 3			-31.37 [-7.87]				-20.50 [-4.72]
Anomaly 4				-2.95 [-0.55]			45.91 [4.63]
Anomaly 5					31.54 [7.55]		20.24 [4.07]
Anomaly 6						-0.84 [-0.13]	15.49 [2.03]
mkt	5.15 [2.04]	5.30 [2.19]	-6.24 [-2.04]	5.94 [2.34]	6.29 [2.58]	5.89 [2.32]	-4.31 [-1.48]
smb	19.74 [5.33]	17.64 [4.93]	-1.83 [-0.42]	19.15 [5.12]	5.56 [1.39]	19.19 [5.08]	-5.69 [-1.29]
hml	-15.78 [-3.20]	-2.54 [-0.50]	4.42 [0.79]	-17.95 [-3.60]	-0.31 [-0.06]	-18.30 [-3.70]	21.71 [3.75]
rmw	35.04 [7.11]	16.55 [3.11]	34.77 [6.36]	35.86 [7.18]	18.55 [3.49]	36.16 [7.27]	9.73 [1.51]
cma	-37.04 [-3.98]	-55.72 [-7.92]	-44.59 [-5.33]	-56.40 [-6.14]	-66.97 [-9.39]	-58.41 [-5.49]	-71.33 [-6.68]
umd	2.74 [1.09]	1.80 [0.74]	9.02 [3.36]	2.23 [0.88]	2.13 [0.88]	2.28 [0.90]	8.08 [3.19]
# months	684	684	504	684	684	684	504
$\bar{R}^2(\%)$	36	40	40	35	40	34	48

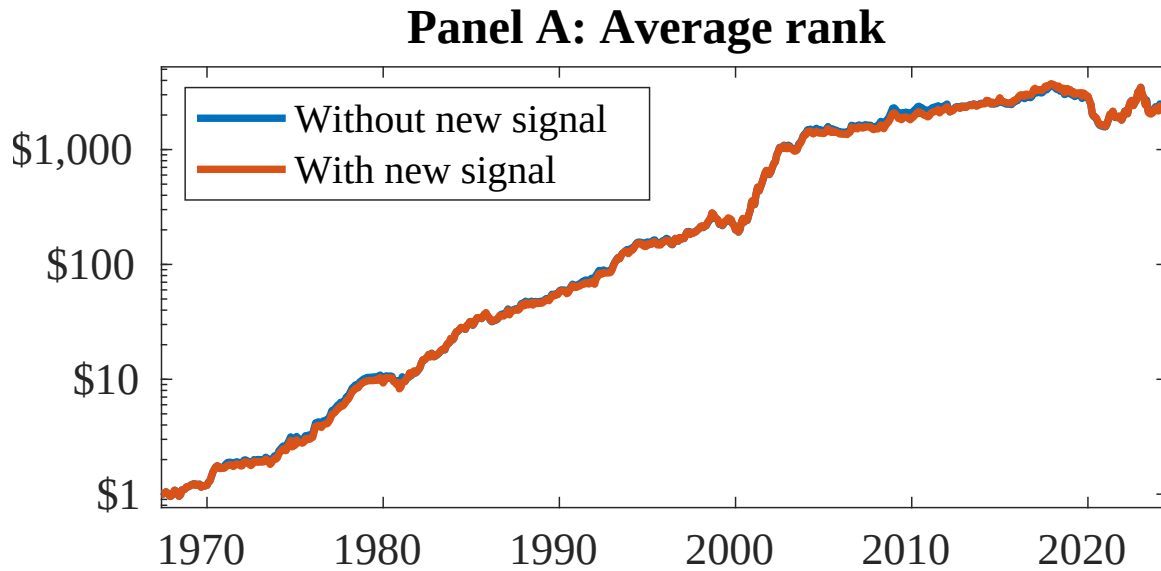


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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